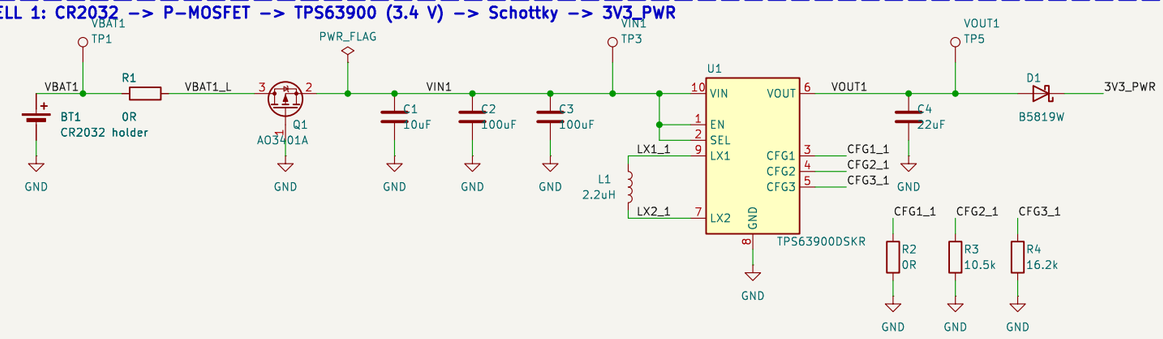
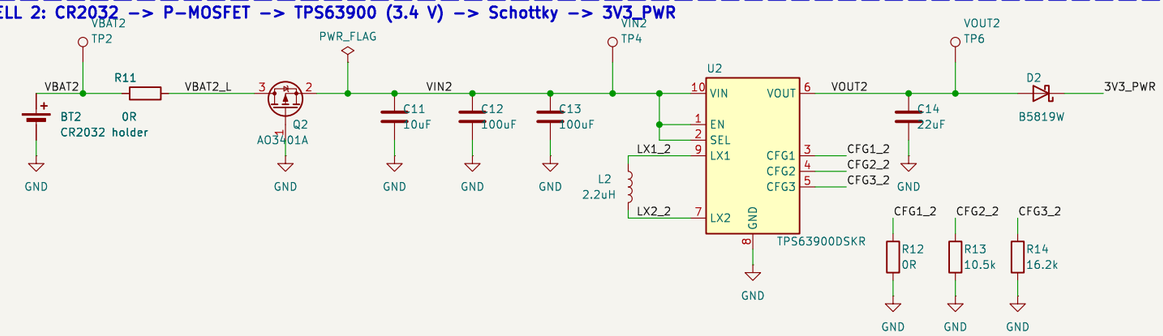
CELL 1: CR2032 -> P-MOSFET -> TPS63900 (3.4 V) -> Schottky -> 3V3_PWR



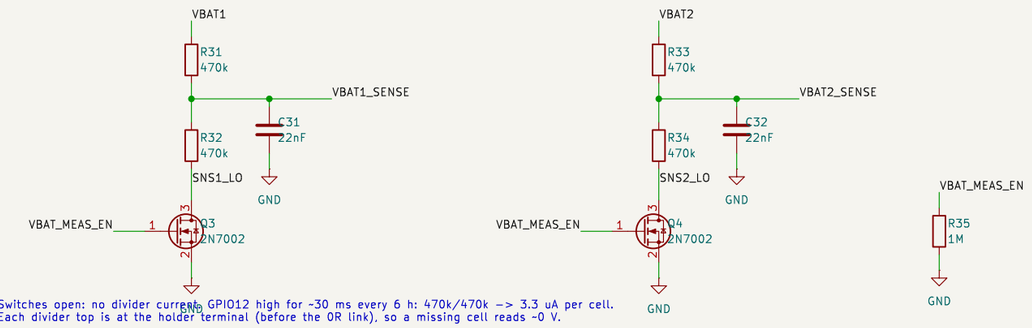
SEL = H: VO(2) = 3.4 V (RCFG1 0R), input limit 100 mA (RCFG2 10.5k; 13.3k = 50 mA option).
VO(1) = 3.3 V (RCFG3 16.2k, unused). TI SLVSET3D tables 7-2/7-3, +-1 % resistors.

CELL 2: CR2032 -> P-MOSFET -> TPS63900 (3.4 V) -> Schottky -> 3V3_PWR



SEL = H: VO(2) = 3.4 V (RCFG1 0R), input limit 100 mA (RCFG2 10.5k; 13.3k = 50 mA option).
VO(1) = 3.3 V (RCFG3 16.2k, unused). TI SLVSET3D tables 7-2/7-3, +-1 % resistors.

CELL VOLTAGE SENSE (switched 470k/470k dividers, ADC1_CH0 / ADC1_CH3)



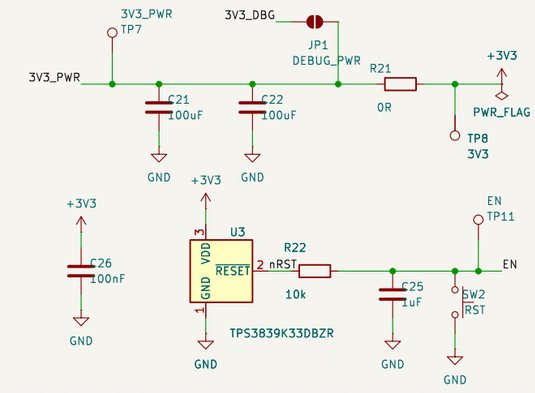
Switches open: no divider current. GPIO12 high for ~30 ms every 6 h: 470k/470k -> 3.3 uA per cell.
Each divider top is at the holder terminal (before the OR link), so a missing cell reads ~0 V.

TEST PADS, MECHANICAL, NOTES

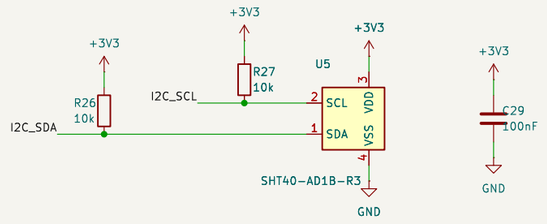


CR2032 ONLY - NOT RECHARGEABLE. Both cells must be the same brand/lot and replaced together.
Power-path decision and its limits: docs/decisions.md, docs/power_budget.md (ESTIMATE until measured).
Current measurement: remove R1 / R11 (per cell) or R21 (load) and insert the ammeter.
U3 TPS3839K33: 2.93 V typ (2.857-2.974 V), push-pull ~RESET, 200 ms delay; R22 keeps it from fighting SW2 / TP11 (EN).

3V3 RAIL, SUPERVISOR, RESET, DEBUG SUPPLY

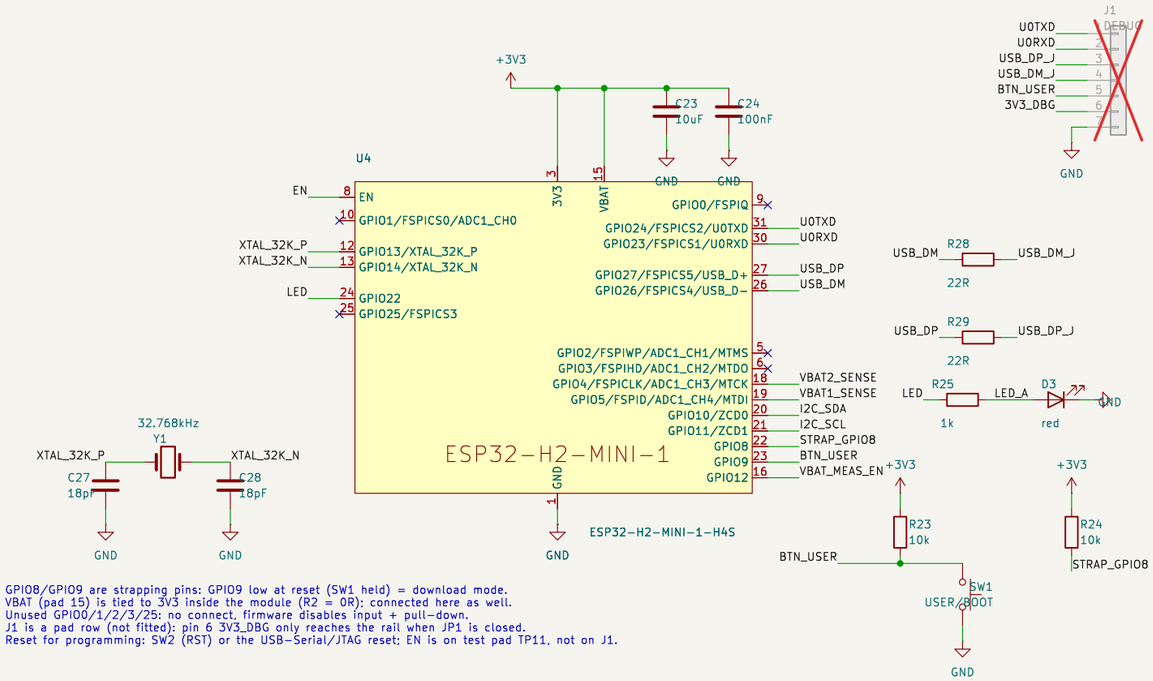


SHT40 (on a slotted tongue, away from converters and module)



No copper under the sensor except its pads; do not solder the die pad (SHT4x datasheet).

ESP32-H2-MINI-1-H4S, 32.768 kHz CRYSTAL, USER/BOOT BUTTON, LED, DEBUG PADS



GPIO8/GPIO9 are strapping pins: GPIO9 low at reset (SW1 held) = download mode.
VBAT (pad 15) is tied to 3V3 inside the module (R2 = 0R): connected here as well.
Unused GPIO0/1/2/3/25: no connect, firmware disables input + pull-down.
J1 is a pad row (not fitted): pin 6 3V3_DBG only reaches the rail when JP1 is closed.
Reset for programming: SW2 (RST) or the USB-Serial/JTAG reset; EN is on test pad TP11, not on J1.

Generated by hardware/scripts/gen_schematic.py from design.py - edit there
ERC/DRC reports: hardware/reports
Power path and battery life figures are ESTIMATES (docs/power_budget.md)
ESP32-H2 Matter over Thread, SHT40, 2 x CR2032 (one TPS63900 per cell)

TH-SENSOR-H2

Sheet: /
File: th_sensor_h2.kicad_sch

Title: TH-SENSOR-H2 thermo-hygrometer

Size: A3	Date: 2026-10-29	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1